Computational Physics WS 2026/27 – Assignment 1

Julian Köldorfer

Due: 01.11.2026

Exercise 1a)

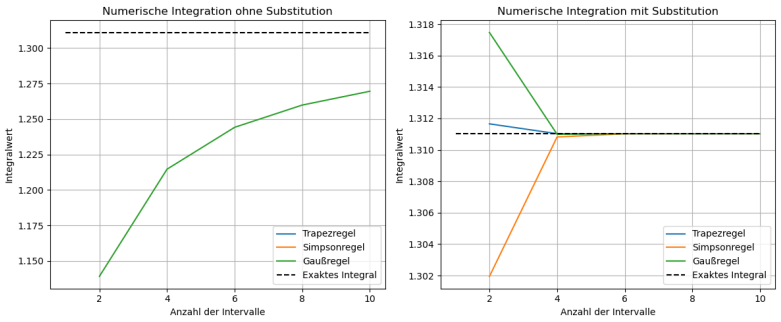


Figure 1: Exercise 1a): integral value versus number of intervals for the trapezoidal rule, Simpson's rule and Gauss quadrature, without (left) and with (right) substitution.

The integral $F = \int_0^1 \mathrm{d}y/\sqrt{1-y^4}$ was computed with the trapezoidal rule, Simpson's rule and Gauss quadrature for $N = 2, 4, \ldots, 10$ intervals (nodes), first directly and then with the substitution $y = \sin u$, which removes the singularity at $y = 1$ ($F = \int_0^{\pi/2} \mathrm{d}u/\sqrt{1+\sin^2 u}$). Fig. 1 compares the integral value of the three methods with the exact value as a function of N.

Exercise 1b)

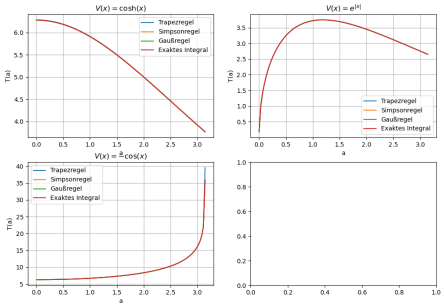


Figure 2: Exercise 1b): period $T(a)$ for $\cosh x$, $e^{|x|}$ and $-\cos x$.

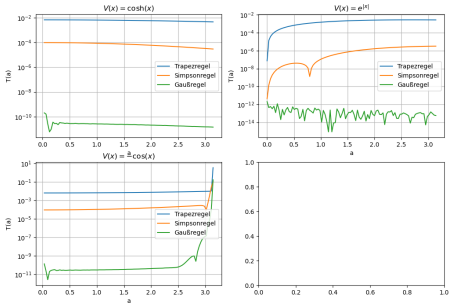


Figure 3: Exercise 1b): deviation of the numerical integrals from the exact value (log scale).

The period $T(a) = \sqrt{8m}\int_0^a \mathrm{d}x/\sqrt{V(a)-V(x)}$ was computed for $V(x) = \cosh x$, $e^{|x|}$ and $-\cos x$ with $0 < a < \pi$. The substitution $x = a(1-u^2)$ removes the square-root singularity at the turning point; at $u = 0$ the integrand has the finite limit $2\sqrt{a/V'(a)}$. Fig. 2 shows $T(a)$ for the three methods and the exact integral, Fig. 3 the deviation from the exact value on a logarithmic scale. High precision is needed when $V'(a) \to 0$, e.g. for $-\cos x$ with $a \to \pi$, where $T(a)$ diverges.

Exercise 1c)

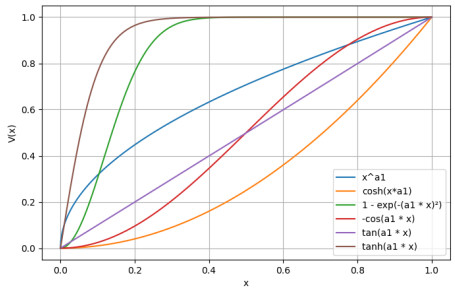


Figure 4: Exercise 1c): normalized potentials ($V(0) = 0$, $V(a) = 1$) for the best a_1 of each family.

$V(x)$	best a_1	$T(a)/T(a/2)$
x^{a_1}	0.5000	2.0000
$\cosh(a_1 x)$	0.1000	1.9997
$1 - e^{-(a_1 x)^2}$	6.0000	294 748.43
$-\cos(a_1 x)$	3.1406	4.7924
$\tan(a_1 x)$	0.1000	1.9971
$\tanh(a_1 x)$	10.0000	143.10

Table 1: Best a_1 and corresponding ratio.

We look for a potential that maximizes $T(a)/T(a/2)$ for $a = 1$ and $V(a) - V(0) = 1$. Families $V(x, a_1)$ were used, normalized such that $V(0) = 0$ and $V(a) = 1$. The shape parameter a_1 was scanned over a range and the maximum of the ratio was determined (Table 1); Fig. 4 shows the resulting potentials. Soft, saturating potentials (Gaussian well, tanh, $-\cos$) give the largest ratios, and the maxima lie at the edge of the scanned a_1 range. The limit is set by numerical precision (Gaussian well, tanh) or by $V'(a) \to 0$ ($-\cos$, $a_1 < \pi$).

Exercise 2a) and 2b)

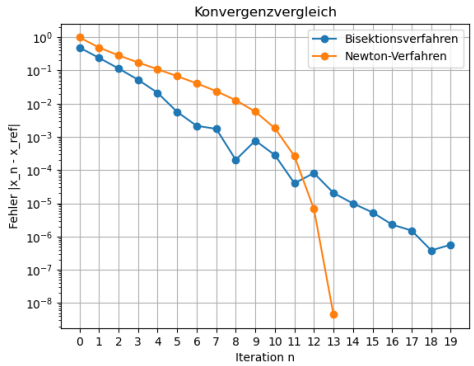


Figure 5: Exercise 2a): error of bisection and Newton versus iteration (log scale).

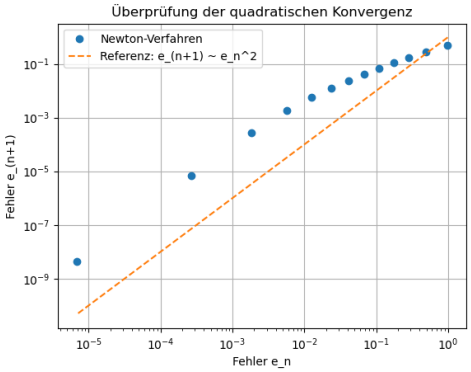


Figure 6: Exercise 2b): e_{n+1} versus e_n with reference e_n^2 (log-log).

The position of the Lagrange point L1 in the Earth–Sun system ($\mu = M_E/(M_S + M_E)$, $R = 1.496 \cdot 10^{11}$ m) is the root of the quintic equation (9). It was found with the bisection method on $[0, 1]$ and with Newton's method (start $x = 1$): $x_{\text{bisection}} = 0.0099688$ and $x_{\text{Newton}} = 0.0099693$, i.e. $r = xR = 1.491 \cdot 10^9$ m $\approx 1.491 \cdot 10^6$ km from the Earth. Fig. 5 shows the error with respect to a high-precision Newton solution used as reference. Bisection converges linearly, Newton's method considerably faster.

To check the quadratic convergence of Newton's method, Fig. 6 plots the error e_{n+1} against e_n on a double-logarithmic scale together with the reference $e_{n+1} \sim e_n^2$. Points parallel to the reference confirm quadratic convergence.

Exercise 3 (bonus)

Polynomials of degree 6 with uniformly distributed coefficients $0 \le a_i \le 1$ were generated. The real roots in $[-10, 10]$ were found via sign changes on a fine grid followed by bisection; roots closer than $\epsilon = 10^{-4}$ count as equal. Over 10 random polynomials the average number of real roots in $[-10, 10]$ is between 0.5 and 1.30.